Scale similarity Matrices and Markov chains

0.1 Scale similarity as Min or Meet Matrices

Matrices $A = \left(a_{\min(i,j)}\right)_{i,j=1}^{N}$, where $a_1, \ldots, a_N$ is a non–negative sequence (in our case: $a_i = f(K_{\varepsilon_i}(S))$), is a well-studied class known as *min matrices* (or, more generally, *meet matrices on posets*). The matrix has a beautiful staircase structure.

0.1.1 $LDL^\top$ Factorisation and Determinant

The matrix admits a clean factorisation $A = LDL^\top$, where L is lower-triangular with ones in each column up to the diagonal, and $D = \operatorname{diag}(d_1, \ldots, d_N)$ with

$$d_1 = a_1, \qquad d_k = a_k - a_{k-1} \quad (k \geq 2).$$

This immediately gives the determinant:

$$\det A \;=\; a_1 \cdot (a_2 - a_1) \cdot (a_3 - a_2) \cdots (a_N - a_{N-1}).$$

0.1.2 Positive (Semi-)Definiteness

From the diagonal D, the matrix A is

- *positive semidefinite* if and only if the sequence $a_1, a_2, \ldots, a_N$ is non-negative and non-decreasing;
- *positive definite* if and only if $0 < a_1 < a_2 < \cdots < a_N$.

0.1.3 Tridiagonal Inverse

When A is invertible (i.e. all $d_k \neq 0$), the inverse A^{-1} is *tridiagonal*. Its entries can be written explicitly in terms of the differences $d_k = a_k - a_{k-1}$.

Setting $d_k = a_k - a_{k-1}$ (with $a_0 := 0$), the inverse A^{-1} has only three nonzero bands.

Diagonal entries:

$$\left(A^{-1}\right)_{ii} \;=\; \begin{cases} \dfrac{1}{d_i} + \dfrac{1}{d_{i+1}} & \text{for } 1 \leq i < N, \\[2ex] \dfrac{1}{d_N} & \text{for } i = N. \end{cases}$$

Off-diagonal entries (superdiagonal and subdiagonal):

$$\left(A^{-1}\right)_{i,\,i+1} \;=\; \left(A^{-1}\right)_{i+1,\,i} \;=\; -\frac{1}{d_{i+1}}, \qquad 1 \leq i < N.$$

All other entries are zero.

In terms of the original sequence a_k:

$$A^{-1} = \begin{pmatrix} \frac{1}{a_1} + \frac{1}{a_2 - a_1} & -\frac{1}{a_2 - a_1} & 0 & \cdots & 0 \\ -\frac{1}{a_2 - a_1} & \frac{1}{a_2 - a_1} + \frac{1}{a_3 - a_2} & -\frac{1}{a_3 - a_2} & \cdots & 0 \\ 0 & -\frac{1}{a_3 - a_2} & \ddots & \ddots & \vdots \\ \vdots & & \ddots & \frac{1}{a_{N-1} - a_{N-2}} + \frac{1}{a_N - a_{N-1}} & -\frac{1}{a_N - a_{N-1}} \\ 0 & \cdots & \cdots & -\frac{1}{a_N - a_{N-1}} & \frac{1}{a_N - a_{N-1}} \end{pmatrix}.$$

Derivation sketch. From $A = LDL^\top$ one obtains $A^{-1} = L^{-\top} D^{-1} L^{-1}$. The key observation is that L^{-1} is *bidiagonal*: it has 1s on the diagonal and -1s on the subdiagonal. Multiplying $L^{-\top} D^{-1} L^{-1}$ then produces the tridiagonal form above.

Remark. The matrix A^{-1} is a discrete Laplacian (second-difference operator) with variable coefficients, which connects directly to the Sturm–Liouville / Brownian motion interpretation of the min matrix.

0.1.4 Eigenvalues $\lambda_1, \ldots, \lambda_N$ of A

From the $LDL^\top$ factorisation with $D = \operatorname{diag}(d_1, \ldots, d_N)$ and $d_k = a_k - a_{k-1}$, the inertia (signs of eigenvalues) of A matches that of D by Sylvester's law. Hence:

- If the sequence a_k is *not* monotonically increasing, some $d_k < 0$, and A has negative eigenvalues.
- Only when $0 < a_1 < a_2 < \cdots < a_N$ are all eigenvalues positive.

Since the hypothesis $a_k \geq 0$ (non-negative but not necessarily increasing) is assumed, the matrix A can be *indefinite*.

0.1.5 Perron–Frobenius Setup

When $a_1 > 0$, the min matrix $A = \left(a_{\min(i,j)}\right)$ has **all strictly positive entries**, so the Perron–Frobenius theorem guarantees:

- a **simple, dominant eigenvalue** $\lambda_1 > 0$ (the spectral radius);
- a corresponding **strictly positive eigenvector** $\mathbf{v} > 0$, unique up to scaling:

$$A\mathbf{v} = \lambda_1 \mathbf{v}, \qquad v_i > 0 \ \ \forall\, i.$$

Since A is symmetric, left and right Perron eigenvectors coincide.

0.1.6 Local Maxima of a_j and v_j Coincide

Starting from the eigenvalue equation $(Av)_i = \lambda_1 v_i$:

$$(Av)_i \;=\; \sum_{j=1}^{i} a_j\, v_j \;+\; a_i \sum_{j=i+1}^{N} v_j \;=\; \lambda_1\, v_i.$$

Subtracting consecutive equations (index $i+1$ minus index i):

$$(Av)_{i+1} - (Av)_i \;=\; a_{i+1} \sum_{j=i+1}^{N} v_j \;-\; a_i \sum_{j=i+1}^{N} v_j \;=\; (a_{i+1} - a_i) \sum_{j=i+1}^{N} v_j.$$

This yields the **fundamental recurrence**:

$$\boxed{\lambda_1(v_{i+1} - v_i) \;=\; d_{i+1} \cdot S_{i+1}, \qquad S_{i+1} \;:=\; \sum_{j=i+1}^{N} v_j,}$$

where $d_{i+1} = a_{i+1} - a_i$.

Proof That Local Maxima Coincide

Since $v_j > 0$ (Perron eigenvector) and $\lambda_1 > 0$, the tail sums satisfy $S_{i+1} > 0$ for all $i < N$. Therefore

$$\operatorname{sign}(v_{i+1} - v_i) \;=\; \operatorname{sign}(d_{i+1}) \;=\; \operatorname{sign}(a_{i+1} - a_i).$$

That is, v and a are **comonotone**: they increase and decrease at exactly the same indices.

Local maximum at an interior point m $(1 < m < N)$.

$$\begin{aligned}
& a \text{ has a local max at } m \\
\iff\; & a_{m-1} < a_m \text{ and } a_m > a_{m+1} \\
\iff\; & d_m > 0 \text{ and } d_{m+1} < 0 \\
\iff\; & v_{m-1} < v_m \text{ and } v_m > v_{m+1} \\
\iff\; & v \text{ has a local max at } m. \qquad \blacksquare
\end{aligned}$$

The same argument applies to **local minima**, **monotone segments**, and **boundary extrema** ($i = 1$ or $i = N$). The complete monotonicity profile of a and v is identical.

Global Maximum: Does Not Necessarily Coincide

The global maxima can differ. The mechanism is clear from the identity

$$v_m - v_j \;=\; \frac{1}{\lambda_1} \sum_{k=j+1}^{m} d_k\, S_k.$$

While $a_m - a_j = \sum_{k=j+1}^{m} d_k$, the eigenvector version weights each d_k by S_k. Since S_k is **decreasing in** k (a tail sum of positive terms), earlier differences are amplified more, creating a systematic bias:

- Increases **early** in the sequence are **amplified** (large S_k).
- Increases **late** in the sequence are **damped** (small S_k).

Counterexample Construction

Take a sequence with two local maxima: an early moderate peak and a late large peak,

$$a \;=\; (1,\; 10,\; 1,\; 1,\; 1,\; 1,\; 1,\; 1,\; 1,\; 12), \qquad N = 10.$$

The global max of a is at $j = 10$ ($a_{10} = 12 > a_2 = 10$). Comparing the two decisive jumps:

- **Jump to the early peak** ($i = 1 \to 2$): $d_2 = 9$, weighted by $S_2 = \sum_{j=2}^{10} v_j$ (9 terms) — *large* weight.
- **Recovery to the late peak** ($i = 9 \to 10$): $d_{10} = 11$, weighted by $S_{10} = v_{10}$ (single term) — *small* weight.

Between indices 3 and 9, $d_k = 0$, so v is flat there. The massive amplified jump at $k = 2$ creates a tall early peak in v, while the large but barely amplified jump at $k = 10$ creates a smaller late peak. The result: $v_2 > v_{10}$ even though $a_2 < a_{10}$.

Precise Condition for Global Maximum Shift

Let $m_1 < m_2$ be two local maxima of a with $a_{m_1} < a_{m_2}$ (so m_2 is the global max of a). Then $v_{m_1} > v_{m_2}$ (global max of v shifts to m_1) whenever

$$\sum_{k=m_1+1}^{m_2} d_k\, S_k \;<\; 0,$$

despite $\sum_{k=m_1+1}^{m_2} d_k = a_{m_2} - a_{m_1} > 0$. This happens when the **descent from** m_1 (negative d_k, large S_k) outweighs the **ascent to** m_2 (positive d_k, small S_k) in the S_k-weighted sum.

The Discount-Factor Analogy

The tail sum S_k acts as a **discount factor**: later changes matter less. This is analogous to **net present value** in finance — a unit increase at an early index is worth more (in the eigenvector metric) than the same unit increase at a late index. Consequently, v is biased toward early peaks, and the global maximum of v can sit at an earlier local maximum that is *not* the global maximum of a.

0.1.7 Trace Formulas

The factorisation gives immediate access to traces:

$$\operatorname{tr}(A) \;=\; \sum_{i=1}^{N} a_i \;=\; \sum_{k} \lambda_k, \qquad \operatorname{tr}(A^2) \;=\; \sum_{i,j} a^2_{\min(i,j)} \;=\; \sum_{k} \lambda_k^2.$$

The second trace can be computed via the Volterra structure:

$$\operatorname{tr}(A^2) \;=\; \operatorname{tr}(LDL^\top LDL^\top) \;=\; \operatorname{tr}(D\,L^\top L\,D\,L^\top L).$$

Since $(L^\top L)_{kl} = N - \max(k,l) + 1$ (itself a "max matrix"), one can evaluate traces in closed form, yielding **sum rules** constraining the eigenvalues.

0.1.8 Perturbation Theory

Since $A = LDL^\top$ and D depends linearly on the a_k, perturbations $a_k \to a_k + \varepsilon_k$ produce

$$\delta A \;=\; L\,(\delta D)\,L^\top, \qquad \delta D \;=\; \operatorname{diag}(\delta d_1, \ldots, \delta d_N), \quad \delta d_k \;=\; \delta a_k - \delta a_{k-1}.$$

This rank-structured perturbation allows efficient eigenvalue tracking via **first-order perturbation theory**:

$$\delta\lambda_j \;=\; \mathbf{u}_j^\top\, L\,(\delta D)\,L^\top\,\mathbf{u}_j \;=\; \sum_k \delta d_k\,(L^\top \mathbf{u}_j)_k^2,$$

where $\mathbf{u}_j$ is the j-th eigenvector. Each d_k influences λ_j proportionally to the squared weight $(L^\top \mathbf{u}_j)_k^2$ — a **cumulative sum of the eigenvector entries**.

0.1.9 Connection to Brownian Motion

The special case $a_k = k$ yields the matrix $\big(\min(i,j)\big)_{i,j}$, which is the covariance matrix of standard Brownian motion (Wiener process) sampled at integer times $t = 1, 2, \ldots, N$. More generally, for $a_k = t_k$ with $0 < t_1 < \cdots < t_N$, this is the covariance matrix of a Wiener process at times t_k.

For $a_k = k$ (the pure $\min(i,j)$ matrix), the eigenvalues are

$$\lambda_k \;=\; \frac{1}{2\big(1 - \cos\theta_k\big)}, \qquad k = 1, \ldots, N,$$

where θ_k are related to roots of certain trigonometric equations. For general a_k, the eigenvalue problem connects to Sturm–Liouville theory through the tridiagonal structure of the inverse.

0.1.10 Meet Matrices on Posets and GCD Matrices

The min matrix is a special case of the *meet matrix* $\big(f(x_i \wedge x_j)\big)$ on a meet-semilattice. The general theory (developed by Lindqvist, Mattila, Haukkanen, and others) encompasses:

- **GCD matrices** $\big(f(\gcd(i,j))\big)$: since gcd is the meet on the divisibility poset, Smith's classical determinant formula

  $$\det\big(\gcd(i,j)\big)_{i,j=1}^{N} \;=\; \prod_{k=1}^{N} \varphi(k)$$

  (Euler's totient) is a special case of the same structural phenomenon.

- **LCM matrices**, which arise as join matrices on the same poset.

The general theory provides determinant formulas, invertibility criteria, and definiteness conditions on arbitrary posets via the Möbius function of the poset.

Key References

- H. J. S. Smith (1876) — the original determinant result for GCD matrices.
- P. Haukkanen, J. Wang, J. Sillanpää (1997) — "On Smith's determinant"; systematic treatment of meet and join matrices on posets.
- M. Mattila and P. Haukkanen — extensive work on meet matrices, their inverses, factorisations, and definiteness on general lattices.
- R. Bhatia (2007) — *Positive Definite Matrices*; discusses min matrices in the context of infinitely divisible matrices and Brownian motion covariances.

0.2 Min Matrices and Markov Chains

0.2.1 Natural Markov Chain from Matrix A

Doob h-Transform (Ground State Transform)

This construction directly uses the **Perron eigenvector v**. Define

$$\widetilde{P} \;=\; \frac{1}{\lambda_1}\, D_v^{-1}\, A\, D_v, \qquad \widetilde{P}_{ij} \;=\; \frac{a_{\min(i,j)}\, v_j}{\lambda_1\, v_i},$$

where $D_v = \mathrm{diag}(v_1, \ldots, v_N)$.

This is stochastic because

$$\widetilde{P}\,\mathbf{1} \;=\; \frac{1}{\lambda_1} D_v^{-1} A\,\mathbf{v} \;=\; \frac{1}{\lambda_1} D_v^{-1}(\lambda_1 \mathbf{v}) \;=\; \mathbf{1}. \quad \checkmark$$

Stationary distribution. Detailed balance holds with

$$\boxed{\tilde{\pi}_i \;=\; \frac{v_i^2}{\sum_k v_k^2}.}$$

Verification. $\tilde{\pi}_i \widetilde{P}_{ij} = \dfrac{v_i^2}{\|\mathbf{v}\|^2} \cdot \dfrac{A_{ij}\, v_j}{\lambda_1\, v_i} = \dfrac{v_i\, A_{ij}\, v_j}{\lambda_1 \|\mathbf{v}\|^2}$, which is symmetric in i, j since $A_{ij} = A_{ji}$.

Direct spectral connection to A. The eigenvalues of $\widetilde{P}$ are exactly

$$\widetilde{\mu}_k \;=\; \frac{\lambda_k}{\lambda_1} \in [-1, 1], \qquad k = 1, \ldots, N,$$

where $\lambda_1 \geq |\lambda_2| \geq \cdots \geq |\lambda_N|$. Some eigenvalues of $\widetilde{P}$ lie in $(-1, 0)$ — these correspond to *oscillatory modes* in the chain's relaxation, where the chain alternates before settling to equilibrium.

The h-Transform via Volterra Structure

Recall the h-transformed chain $\widetilde{P} = \frac{1}{\lambda_1} D_v^{-1} A\, D_v$. Using $A = LDL^\top$:

$$\widetilde{P} \;=\; \frac{1}{\lambda_1}\, D_v^{-1}\, L\, D\, L^\top\, D_v.$$

This reveals $\widetilde{P}$ as a **five-stage process**:

1. **Weight** the state by v_j (via D_v),
2. **Integrate forward** (apply $L^\top$: cumulate),
3. **Reweight** by d_k (apply D),
4. **Integrate backward** (apply L: cumulate again),
5. **Normalise** by v_i and λ_1.

The transition mechanism is thus a **discrete convolution through a cumulation–reweighting–cumulation** pipeline.

Spectral Gap

The Perron–Frobenius eigenvalue $\lambda_1 = \rho(A) > 0$ is still the largest in absolute value (assuming $a_1 > 0$), but the remaining eigenvalues λ_k can be negative. The convergence bound for the h-transformed chain should therefore read

$$\left\|\widetilde{P}^n(i,\cdot) - \tilde{\pi}\right\|_{\mathrm{TV}} \;\le\; C\left(\frac{\lambda^*}{\lambda_1}\right)^n,$$

where the subdominant eigenvalue is

$$\boxed{\lambda^* \;=\; \max_{k\ge 2} |\lambda_k| = |\lambda_2|,}$$

and the spectral gap is

$$\gamma \;=\; 1 - \frac{\lambda^*}{\lambda_1}.$$

The ratio λ^*/λ_1 can be driven by either the second-largest positive eigenvalue or the most negative eigenvalue, depending on the sequence a_k.

0.2.2 Connection to Birth–Death Chains

Let the sequence a_i, $i = 1, \dots, N$ be monotonously increasing. Since A^{-1} is **tridiagonal**, define the discrete Laplacian

$$\mathcal{L} \;=\; -A^{-1},$$

which has negative diagonal entries, positive off-diagonal entries $1/d_{i+1}$ connecting neighbours $i \leftrightarrow i+1$, and (nearly) zero row sums for interior states. Hence $\mathcal{L}$ acts as the **generator of a continuous-time birth–death chain** on $\{1, \dots, N\}$, with transition rates

$$q_{i\to i+1} \;=\; \frac{1}{d_{i+1}} \;=\; \frac{1}{a_{i+1} - a_i}, \qquad q_{i\to i-1} \;=\; \frac{1}{d_i} \;=\; \frac{1}{a_i - a_{i-1}}.$$

The **birth and death rates are inversely proportional to the gaps** $a_{i+1} - a_i$: where the a_k are closely spaced, transitions are fast; where they are widely spaced, transitions are slow.

The corresponding discrete-time birth–death chain is

$$P^{\mathrm{BD}}_{i,\,i+1} \;=\; \frac{1/d_{i+1}}{1/d_i + 1/d_{i+1}}, \qquad P^{\mathrm{BD}}_{i,\,i-1} \;=\; \frac{1/d_i}{1/d_i + 1/d_{i+1}},$$

for interior states, with appropriate boundary modifications.

The deep picture is: **the min matrix simultaneously encodes a dense reversible chain (via its entries) and a nearest-neighbour birth–death chain (via its tridiagonal inverse), both connected through the Perron eigenvalue and eigenvector of A.**

0.3 Min Matrices as Discrete Volterra/Hardy Operators

The Factorisation $A = LDL^\top$

Define the **discrete integration (Volterra) operator** L as the lower-triangular matrix

$$L_{ij} \;=\; \begin{cases} 1 & \text{if } j \le i, \\ 0 & \text{otherwise,} \end{cases} \qquad (Lf)(i) \;=\; \sum_{k=1}^{i} f(k),$$

and let $D = \mathrm{diag}(d_1, \ldots, d_N)$ with $d_k = a_k - a_{k-1}$. Then

$$(LDL^\top)_{ij} \;=\; \sum_{k=1}^{i}\sum_{l=1}^{j} d_k\,\delta_{kl} \;=\; \sum_{k=1}^{\min(i,j)} d_k \;=\; a_{\min(i,j)} \;=\; A_{ij},$$

giving the core identity

$$\boxed{A \;=\; L\,D\,L^\top.}$$

The min matrix is the **composition of a discrete Volterra operator, a diagonal weight, and the adjoint Volterra operator**. When $d_k = 1$ for all k (i.e. $a_k = k$), this simplifies to $A = LL^\top$ — the **Gram matrix** of L.

Continuous Analogue: The Volterra Operator

The matrix L is the discrete analogue of the **Volterra integral operator**

$$(Vf)(x) \;=\; \int_0^x f(t)\,dt.$$

The corresponding continuous min-kernel is

$$(VV^*g)(x) \;=\; \int_0^1 \min(x,y)\,g(y)\,dy,$$

which is the **covariance operator of Brownian motion** on $[0,1]$. The weighted version becomes

$$(V\,\sigma\,V^*g)(x) \;=\; \int_0^1 \left(\int_0^{\min(x,y)} \sigma(t)\,dt\right) g(y)\,dy,$$

corresponding to the weighted min matrix with $\sigma(t)\,dt$ replacing d_k.

Key spectral fact about V. The Volterra operator is **compact and quasinilpotent** — its spectrum is $\{0\}$, yet it has no eigenvalues. However, VV^* is **self-adjoint and positive**, with eigenvalues

$$\lambda_k(VV^*) \;=\; \frac{1}{\pi^2\big(k-\frac{1}{2}\big)^2}, \qquad k = 1, 2, 3, \ldots$$

This gives the **asymptotic eigenvalue distribution** for large min matrices.

Connection to Hardy Operators

The **discrete Hardy operator** is

$$(Hf)(i) \;=\; \frac{1}{i}\sum_{k=1}^{i} f(k),$$

which equals $H = \Lambda^{-1}L$ with $\Lambda = \mathrm{diag}(1, 2, \ldots, N)$. Hence

$$A \;=\; L\,D\,L^\top \;=\; (\Lambda H)\,D\,(\Lambda H)^\top \;=\; \Lambda\,H\,D\,H^\top\,\Lambda.$$

Hardy's inequality states that for $p > 1$,

$$\sum_{i=1}^{\infty}\left(\frac{1}{i}\sum_{k=1}^{i} f(k)\right)^p \;\le\; \left(\frac{p}{p-1}\right)^p \sum_{i=1}^{\infty} |f(i)|^p,$$

equivalently the operator norm bound $\|H\|_{\ell^p\to\ell^p} \le p/(p-1)$. For $p = 2$:

$$\|H\|_{\ell^2\to\ell^2} \;\le\; 2, \qquad \text{hence} \quad \|L\|^2 = \|LL^\top\| = \lambda_1(A) \;\le\; \|H\|^2 \cdot N^2 \quad \text{(with appropriate scaling)}.$$

Mixing Time Bounds via Hardy Inequalities

Hardy's inequality controls $\|L\|$, which bounds $\|A\| = \|LDL^\top\| \le \|L\|^2\|D\|$, giving

$$\lambda_1 \;\le\; \|L\|^2 \cdot \max_k |d_k|.$$

Combined with lower bounds (e.g. from $\mathrm{tr}(A)$), one obtains **spectral gap estimates**:

$$\gamma \;=\; 1 - \frac{\lambda^*}{\lambda_1} \;\ge\; 1 - \frac{\|L\|^2 \cdot \max|d_k|}{\mathrm{tr}(A)/N} \cdot \frac{1}{N}.$$

In the monotone case, sharper weighted Hardy inequalities translate directly into **mixing time bounds** for the associated chain.

Computing $\|L\|^2$ Explicitly

Since $\|L\|^2 = \|LL^\top\| = \lambda_{\max}(LL^\top)$, and

$$(LL^\top)_{ij} \;=\; \sum_{k=1}^{N} \mathbf{1}_{k\le i}\,\mathbf{1}_{k\le j} \;=\; \min(i,j),$$

we need the largest eigenvalue of the pure min matrix $M_{ij} = \min(i,j)$, which has $d_k = 1$ for all k.

Finding All Eigenvalues of M

Since $M^{-1} = \Delta^\top\Delta$ (with $D = I$), the inverse is

$$M^{-1} = \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 1 \end{pmatrix}.$$

This is the discrete Laplacian with a Dirichlet condition at 0 and a Neumann condition at N.

The eigenvector ansatz $u_i = \sin(i\theta)$ satisfies $u_0 = 0$ (Dirichlet) and the bulk equation. The Neumann-type condition at $i = N$ requires

$$\sin\big((N+1)\theta\big) - \sin(N\theta) \;=\; 0,$$

which gives $\cos\left(\dfrac{(2N+1)\theta}{2}\right) = 0$, and hence

$$\theta_k \;=\; \frac{(2k-1)\pi}{2N+1}, \qquad k = 1,\dots,N.$$

The eigenvalues of M^{-1} are therefore

$$\mu_k \;=\; 2 - 2\cos\theta_k \;=\; 4\sin^2\left(\frac{(2k-1)\pi}{2(2N+1)}\right),$$

and consequently the eigenvalues of $M = LL^\top$ are

$$\boxed{\lambda_k(M) \;=\; \frac{1}{4\sin^2\left(\dfrac{(2k-1)\pi}{2(2N+1)}\right)}, \qquad k = 1,\dots,N.}$$

The Operator Norm

The largest eigenvalue corresponds to $k = 1$:

$$\boxed{\|L\|^2 \;=\; \lambda_1(M) \;=\; \frac{1}{4\sin^2\!\Big(\dfrac{\pi}{2(2N+1)}\Big)}.}$$

Asymptotics for Large N

Using $\sin(x) \approx x$ for small x:

$$\|L\|^2 \;\approx\; \frac{1}{4\cdot\dfrac{\pi^2}{4(2N+1)^2}} \;=\; \frac{(2N+1)^2}{\pi^2} \;\approx\; \frac{4N^2}{\pi^2}.$$

More precisely,

$$\|L\|^2 \;=\; \frac{4N^2}{\pi^2}\left(1+\frac{2}{N}+O(N^{-2})\right).$$

For general k,

$$\lambda_k(M) \;\approx\; \frac{4N^2}{\pi^2(2k-1)^2}, \qquad N\to\infty,$$

confirming the k^{-2} decay inherited from the Volterra structure.

Connection to the Continuous Volterra Operator

The continuous Volterra operator $(Vf)(x) = \int_0^x f(t)\,dt$ on $L^2[0,1]$ satisfies

$$\|V\|^2 \;=\; \|VV^*\| \;=\; \frac{1}{\pi^2/4} \;=\; \frac{4}{\pi^2}.$$

The discrete result is entirely consistent: the operator L acts on $\ell^2(\{1,\dots,N\})$ and the scaling $\frac{1}{N}L \approx V$ gives

$$\|L\|^2 \;\approx\; N^2\|V\|^2 \;=\; \frac{4N^2}{\pi^2}.$$

The exact formula $\lambda_1(M) = \big[4\sin^2\pi/(2(2N+1))\big]^{-1}$ refines this with finite-N corrections.

0.3.1 Eigenvalue Asymptotics

Let a_j be a monotonously increasing sequence.

The factorisation $A = LDL^\top$ allows one to analyse the spectrum of A through the **singular values of** $L\sqrt{|D|}$ (when $D > 0$). In the large-N limit with $a_k \approx a(k/N)$ for a smooth function $a(\cdot)$, the eigenvalues of $\frac{1}{N}A$ converge to those of the weighted Volterra integral operator $V\,a'\,V^*$, and one obtains the Weyl-type asymptotics

$$\lambda_k \;\sim\; \frac{N}{\pi^2\big(k-\frac{1}{2}\big)^2}\cdot\langle a'\rangle,$$

where $\langle a'\rangle$ reflects the average of the weight. The k^{-2} decay is universal and stems from the Volterra structure.

If now $D > 0$ does not hold, the factorisation $A = LDL^\top$ still links the spectrum of A to the singular structure of L, but the analysis splits. Write $D = D^+ - D^-$ where $D^+ = \mathrm{diag}(\max(d_k, 0))$ and $D^- = \mathrm{diag}(\max(-d_k, 0))$. Then

$$A \;=\; \underbrace{LD^+L^\top}_{\text{positive semi-definite}} \;-\; \underbrace{LD^-L^\top}_{\text{positive semi-definite}},$$

a difference of two PSD matrices. The eigenvalues of A are controlled by **interlacing** and **Weyl's inequalities**:

$$\lambda_k(A) \;\geq\; \lambda_k(LD^+L^\top) - \|LD^-L^\top\|, \qquad \lambda_k(A) \;\leq\; \lambda_k(LD^+L^\top) + \|LD^-L^\top\|.$$

The k^{-2} asymptotic decay still governs the positive part $LD^+L^\top$, but the negative part can shift, reorder, or create negative eigenvalues. Hence:

- The **largest** eigenvalue still satisfies $\lambda_1 \sim C\,k^{-2}$-type bounds from the Volterra structure.
- The **tail** eigenvalues can be negative, and their asymptotics depend on the interplay of D^+ and D^-.

0.3.2 Potential Theory and Green's Functions

In the monotone case ($D > 0$), the factorisation gives

$$A^{-1} \;=\; L^{-\top}\,D^{-1}\,L^{-1} \;=\; \Delta^\top\,D^{-1}\,\Delta,$$

where $\Delta = L^{-1}$ is the **discrete difference operator** $(\Delta f)(i) = f(i) - f(i-1)$. Hence $A^{-1} = \Delta^\top D^{-1}\Delta$ is a **discrete divergence-form elliptic operator** $-\nabla\cdot(\sigma\nabla)$ with conductances $\sigma_k = 1/d_k$. This connects to **electrical network theory**:

- $1/d_k$ = conductance of the edge between nodes $k-1$ and k,
- d_k = resistance of that edge,
- A = the Green's function (potential matrix) of the network,
- the **commute time** between states i and j is $\mathbb{E}[\tau_{ij}+\tau_{ji}] = (\text{total weight})\cdot(A^{-1}_{ii} - 2A^{-1}_{ij} + A^{-1}_{jj})$, which by tridiagonality reduces to a **sum of resistances** along the path.

Time-Changed Processes

The factorisation $A = LDL^\top$ admits a **time-change interpretation**. In the continuous setting, if $W(t)$ is Brownian motion, then

$$X(t) \;=\; \int_0^t \sqrt{a'(s)}\,dW(s)$$

is a Gaussian process with covariance $\mathbb{E}[X(s)X(t)] = \int_0^{\min(s,t)} a'(u)\,du = a(\min(s,t))$.

Hence the general min matrix corresponds to a **time-changed Brownian motion**, where $a(\cdot)$ plays the role of the clock. The Markov chain from A is the discrete analogue of this time-changed diffusion, and the Volterra factorisation is the **stochastic integral representation**.